Creating Portrait with Strings

String Art Generation using Radon Transform

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Introduction

Inspired by the string art manual crafts, an idea of algorithmic art generator for creating string art portrait is put forward. The theoretical idea is fitting gradation of light and shadow by density of lines, thus models the image with intersect strings.

Relevant works rely primarily on greedy algorithm, yet Radon Transform is also remarked as an additional approach. [1]

Therefore, the project aims to develop a generator allowing users to apply different models and parameters to create scheme of connecting strings for artistic works on circular canvas.

Method

Radon Transform

Radon transform is a mathematical operation that converts a function defined in two-dimensional space into a set of its integrals over straight lines.

It takes an image and represents it in terms of projections from various angles. Converting the gray level function of origin image to an alpha-offset (α, s) domain function

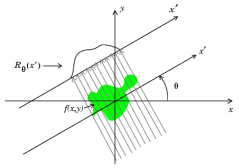


Figure 1. Conversion by integral

This transform is particularly important in medical imaging techniques like computed tomography (CT), where it helps reconstruct cross-sectional images of the body from multiple X-ray projections. [2]

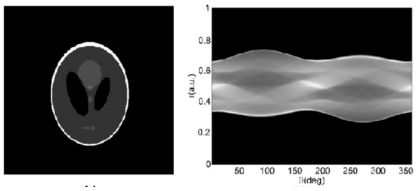


Figure 2. Radon Transform for Medical

In the case of string art generator, Radon transform allows the conversion of an image into a series of line integrals identifying the darkest areas that should be emphasized in the art.

User Interface

The project separate GUI from computation models for clearer and extendable structure. The GUI class view.py invokes algorithms encapsulated in computational classes which takes in customer parameters and output the string scheme for either rendering by GUI or manual craft. Either parameters and scheme stores in JSON format as middleware.

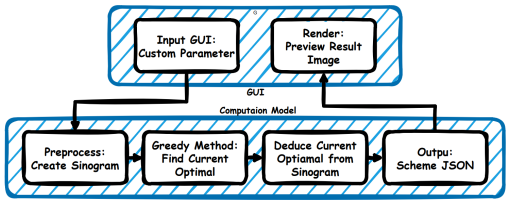


Figure 1. Image transformation process

Computation Model

In the core algorithm, origin image will be preprocessed to enhance its quality and contrast resolution.



Figure 3. Enhance Portrait Contrast

According to greedy algorithm conception, the computational process start by randomly choosing a nail and select the currently best fitting string as one step of string art scheme. Then removed sinogram of optimal string from the sinogram of origin picture for further iteration.



Figure 4. Radon Sinogram of Line

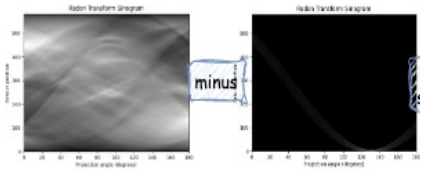


Figure 5. Remove Line Sino from Origin

Results

The algorithm works well with high contrast portrait which allows strings fit different gray level.



Figure 6. Example Result

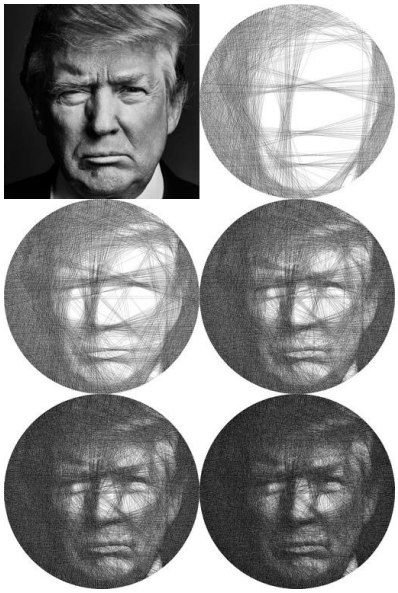


Figure 7. Steps of String Art Generation

However, as expected, the fitness for anime pictures shows lame accuracy due to lower frequency gray levels.

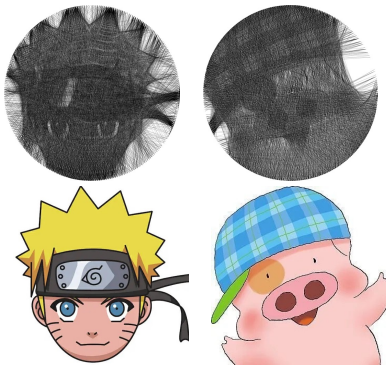


Figure 8. Cartoon and anime picture results

Conclusions

The project created as sophisticated framework as well as eligible computational model for string art generator application. Bringing idea of Radon Transform from medical imaging, the computation performance and outcome image precision is enhanced.

Yet the radon transform model works lamely on anime pictures (low frequency) and exceedingly complex images (high frequency). A possible solution for processing anime pictures would be edge detection. The model could focus on the edge of subjects in the image, then let it fit the edge instead of the color blocks with strings. Hough transform or gradient detection are two eligible approach to do so.[3]

References

[1] Demoussel, at. al. , *A Greedy Algorithm for Generative String Art*, Aug 2022. hai id: hal-03901755. Available: https://hal.science/hal-03901755
[2] M. Zvolsk´y, *Tomographic Image Reconstruction - An Introduction*, Nov. 2014. [Online]. Available: https://www.desy.de/~garutti/LECTURES/BioMedical/Lecture7_ImageReconstruction.pdf
[3] Virtually Passed, *How To Make a Computer Create Something Beautiful: String Art, Part II* of "The Mathematics of String Art", May 2024. [Online]. Available: https://www.youtube.com/watch?v=dBlSmg5T13M.